

# Dimenso Hackathon Report — egocentric lab demo

## → Unitree G1 + Inspire in simulation

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Recover human **hand and arm** motion from smart-glasses head-cam RGB + head IMU, retarget it onto a Unitree **G1 with five-finger Inspire hands**, and replay it in MuJoCo. **Motion translation only** — human demo in, robot joint trajectory out, replayed in sim. No object interaction or manipulation planning. All numbers below are measured from the final, IK-tuned pipeline on `task_04_pasteur_pipette`.

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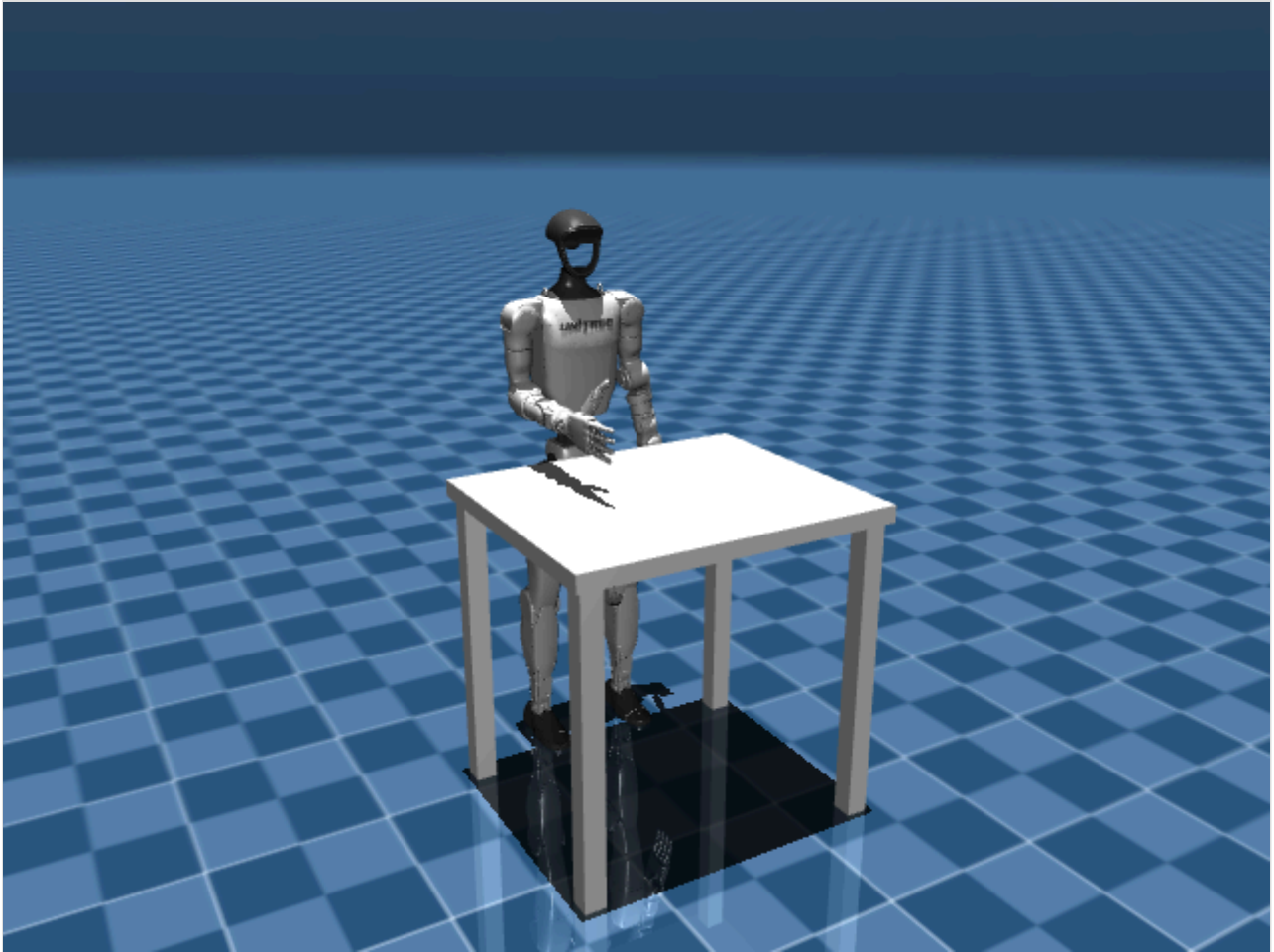
### 1. Problem framing & assumptions

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We chose **task\_04 (Pasteur-pipette)** from the four recorded lab tasks. The choice is grounded in the diagnostics hand-detection sweep ( `docs/diagnostics.md` ): task\_04 had a hand visible in **77.0%** of sampled frames versus 34–49% for the others, the **highest** detection confidence (0.976 vs ~0.95), and the **shortest** no-hand dropout (~3.4 s). Since hand *presence in frame* — not detector quality — is the binding constraint, task\_04 is the most tractable demonstration.

The core reframe: **a forward-facing egocentric camera recovers hands and forearms, not the full body**. The diagnostics Pose check is blunt — only **29% of frames return any MediaPipe Pose result**, and when one fires the lower body is unreliable (knee visibility 0.46, ankle 0.22; the high shoulder/hip "visibility" is the model hallucinating an unseen torso). So we do **not** attempt whole-body capture: we translate hand+arm motion onto the G1's **upper body**, **freeze the legs** at the stand keyframe, and drive the **torso from the head IMU** (the G1 has no neck joint; after tuning the torso stays upright and only **turns** — see §3/§4).

The dexterous motion maps onto a **five-finger Inspire DFQ hand grafted onto the menagerie G1** — one MuJoCo model, **53 actuated DoF** (29 G1 + 24 finger joints), documented in `docs/inspire_hand_model.md`. The robot works over a white table whose top sits at the height the pipeline maps wrists onto (0.75 m):



## 2. Data pipeline (the centerpiece)

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`data_pipeline/run_offline.py` chains five stages; all joint/actuator names are read from the MuJoCo model at run time, nothing hardcoded.

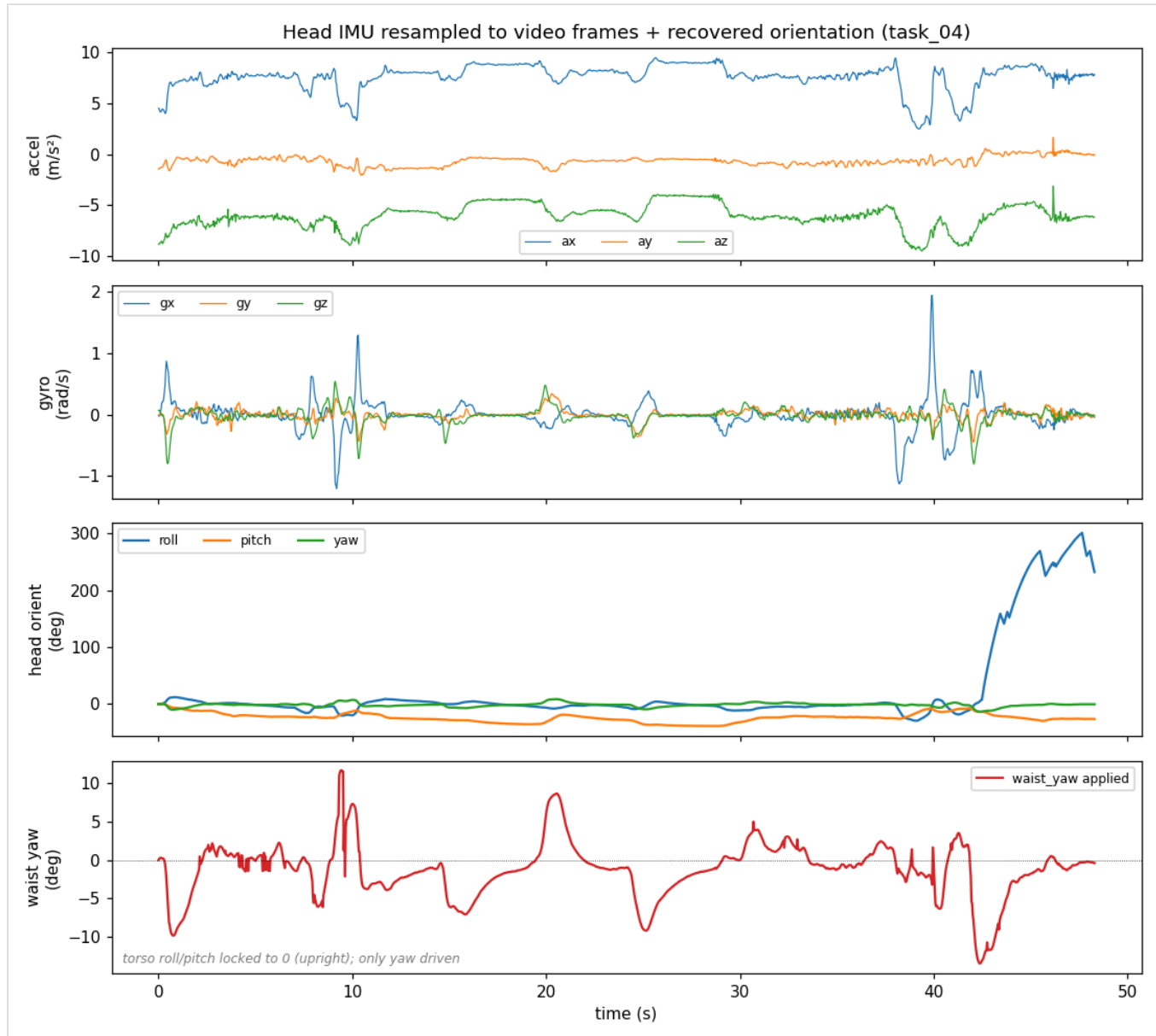
**Perception.** Video decoded with real per-frame timestamps; **MediaPipe Hands** per frame → 21 landmarks/hand + handedness + confidence. Landmarks are de-jittered by a **One Euro filter** (Casiez et al., 2012) and confidence-gated. We derive per-finger **curl** (path-length ratio), the thumb-index **pinch**, and **thumb-tip ↔ fingertip distances** (the descriptor the tuned thumb retargeter uses). Handedness is flipped for the non-mirrored egocentric view.

**Head IMU → waist.** The IMU is ~100 Hz but jittery (per-sample dt 1–18 ms), so we **resample it onto the video frame timestamps** (not a fixed 10 ms grid). A complementary filter recovers head orientation (accel-gravity roll/pitch + gyro integration, yaw leaked), expressed relative to frame 0. The tuned retargeter uses **only the head yaw** to *turn* the waist and keeps the torso upright (§3).

**2D → 3D wrist target (approximated).** RGB-only, no depth, so the target is **approximated**: image position → table-plane X, hand scale (+ MediaPipe z) → depth proxy, and the **lateral Y now tracks the real detected wrist separation**. This is a monotonic guess, not a calibrated reconstruction — the largest single source of error.

**Output — the dataset.** `outputs/task04_dataset.npz` + a committed `task04_dataset.schema.json` document **every** per-frame field, time-aligned to the video frames: frame index/timestamp; the **raw**

**MediaPipe landmark layer** (both hands, 21×3); handedness/confidence/present + stow flags; curl + thumb-distances; the **resampled IMU** (accel, gyro) + recovered head orientation (roll/pitch/yaw) + the **waist-yaw actually applied**; approximate 3D wrist targets; and the **full 53-joint G1+Inspire trajectory** with every joint column named ( `joint_targets` + `actuated_joint_names` , plus full `qpos` ). The `.npz` is gitignored; the schema is the committed contract. An IMU visualization ( `report/figures/imu_visualization.png` ) and sanity plots are produced by `notebooks/01_explore.ipynb` .



### 3. Method (perception → retargeting → imitation learning → simulation)

**Retargeting.** `method/retarget.py` solves **damped-least-squares IK** (MuJoCo body Jacobians) so each detected wrist reaches its 3D target, both arms jointly. After tuning: the **torso stays upright** — `waist_roll` / `waist_pitch` are locked to 0 and **only `waist_yaw`** is driven (softly, by head yaw), so the robot *turns* toward where the person looked but never leans. Fingers use **distance-based retargeting**:

the human thumb ↔ fingertip distance maps to a closeness that flexes both that finger and the **thumb** (so the thumb actually flexes into pinches), with curl as a fallback. When a hand isn't seen, that arm **stows hanging straight down** (out of the egocentric frame) after a hold-last-valid window; all targets are **velocity-clamped** for smooth eases.

**Imitation learning — a sketch (honestly, not built).** This is the one pillar we did not implement, so we reason about it concretely rather than hand-wave.

*Framing.* The dataset is **demonstration data** — per-frame **state** → **action pairs**: observations (proprioception + vision) paired with the commanded joint targets. It is **not** a closed-loop training environment: there is **no depth, no object/scene state, no reward, and it is open-loop** (the robot never acts back on the world). So it can seed imitation learning but cannot, alone, train a *task* policy. We state that plainly.

*Candidate models (inputs → outputs, why):* - **Behavior cloning (BC) MLP/CNN baseline** — obs → next joint targets, supervised. Why: simplest baseline to validate the data pipeline and get a first success-rate number. - **Diffusion Policy** (Chi et al., 2023) — obs → a *distribution* over short action chunks. Why: handles multimodal, contact-rich manipulation far better than a unimodal BC regressor; strong dexterous-manipulation baseline. - **ACT / Action-Chunking Transformer** (ALOHA, Zhao et al., 2023) — obs → chunk of future actions via a CVAE transformer. Why: built for fine bimanual teleop demonstrations, exactly our regime (two arms + dexterous hands). - **VLA (OpenVLA /  $\pi$ -0 class)** — image + language → action. Why: language-conditioned generalization across tasks/objects once we have many demos; the path to "tell the robot what to do," not just replay one motion.

*Inputs / outputs, concretely.* **Inputs:** proprioception (the 53 G1+Inspire joint states) + vision (the egocentric frame + a proposed **wrist camera** for object state) + optional language. **Outputs:** joint-target (or end-effector-target) sequences — exactly what our dataset already stores per frame.

*Data gathering / scaling.* Collect **many more demos** (order **50–200 per task** for a BC baseline; more for diffusion), **add a depth or wrist-mounted RGB-D camera** so the policy can see the manipulated object, use **multiple operators** for diversity, **auto-segment** object-interaction phases, and **convert to LeRobot v2**. Our pipeline already emits LeRobot-friendly per-frame state-action records (named joints + vision + timestamps), so conversion is a formatting step, not a redesign.

*Validation plan.* (1) **held-out demo tracking error** (does the policy reproduce unseen demos?), then (2) **sim rollout success rate** on the task, then (3) **sim-to-real** on hardware. A tiny overfit-one-demo sanity run was **out of scope** in the available time — an honest gap, not a hidden one.

**Simulation.** The combined model loads in MuJoCo with legs frozen at stand; the retargeted trajectory replays as an offscreen mp4 and in a **live native viewer** ( `sim/replay_gl.py --live` ) at real time.

## 4. Validation

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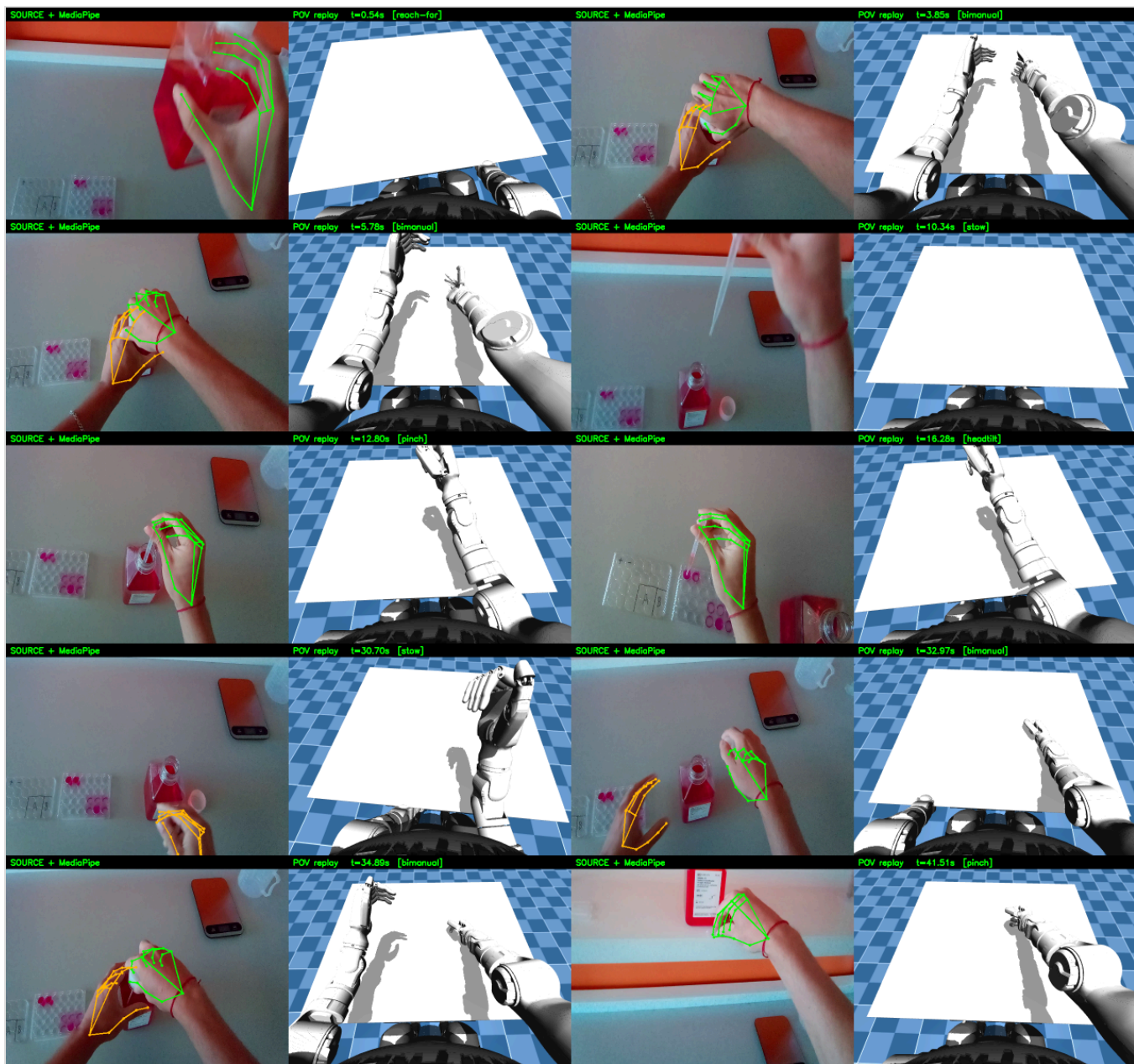
On task\_04 (1431 frames @ 29.6 fps). The tuning pass is reflected here:

- **Tracking vs stow: 79.6% of frames use real tracking, 20.4% stow** (right hand 76.5%, left 26.5%). Stow now **hangs the arm straight down, out of the POV frame**, so a lost hand reads

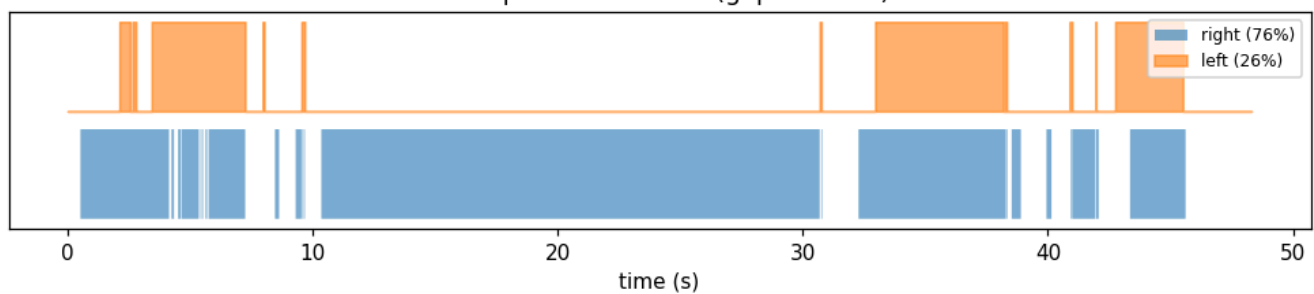
as idle rather than miming an action.

- **Torso upright (tuned):** `waist_roll` and `waist_pitch` are **0.0° on every frame** (was -20... -30° before, a distracting lean); only `waist_yaw` turns, staying within  $\pm 12^\circ$ . This also fixed a latent axis-mismatch (lean `[roll, pitch, yaw]` had been paired against joints `[yaw, roll, pitch]`).
- **Hand separation now tracks the human (tuned):** robot hand-sep went from ~flat ( $\approx 0.29\text{--}0.45$  m, fixed by a `SIDE_BIAS`) to **monotonic with the human** — human 0.326 / 0.375 / 0.384 / 0.555 → robot 0.233 / 0.237 / 0.338 / 0.428.
- **Thumb now flexes into pinches (tuned):** at the pinch frame the thumb went from static (proximal-pitch -0.09, intermediate 0.01, distal 0.01 rad) to **fully flexed ( $\approx 0.60$  /  $0.80$  /  $1.20$  rad)** with the index at 1.70 — previously it only opposed.
- **IMU ↔ video timing:** durations 48.42 s vs 48.30 s ( $\sim 0.12$  s slack), consistent with the documented no-shared-clock assumption.
- **Continuous clip: a 250-frame side-by-side** ( `outputs/tuned_compare_250.mp4` , frames **307–556, 100% right-hand coverage**) plays the source || POV replay frame-aligned; the live viewer plays the same trajectory in real time.

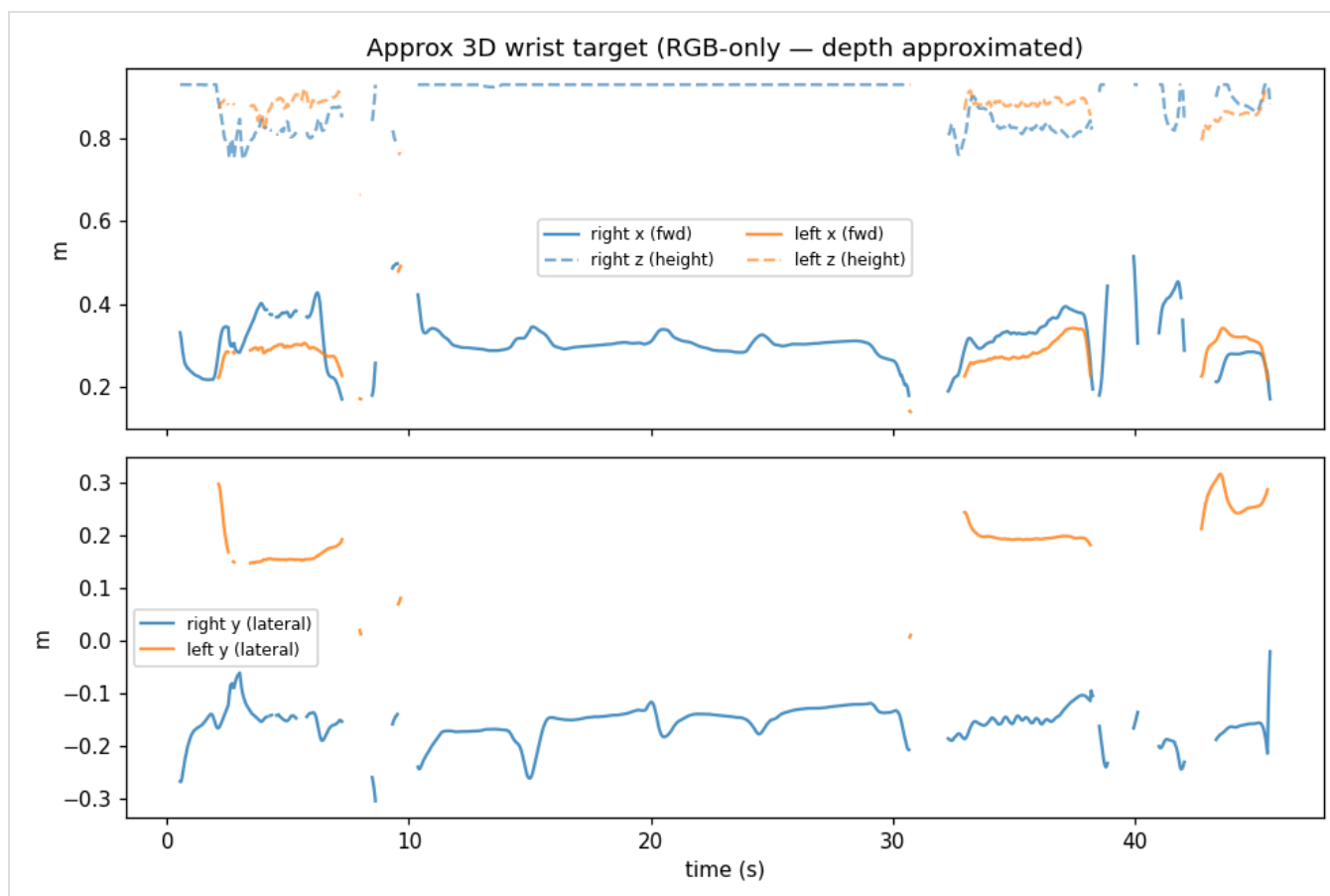
Tuned side-by-side contact sheet (source + MediaPipe || G1+Inspire from the head POV):



Hand-present timeline (gaps → stow)







## 5. Feasibility & cost proposal

*(Proposal estimates with stated assumptions, not measured results.)*

**BOM (research MVP).** Egocentric capture with depth — depth-capable smart glasses (~\$2–3k) or standard glasses + a wrist-mounted RGB-D camera (RealSense-class, ~\$250–400); one **Unitree G1** (~\$16k entry class); **two Inspire five-finger hands** (RH56-class, ~\$3–5k each); one training workstation (RTX-4090-class GPU ~\$1.6k, or cloud A100 at ~\$1.5–3/hr). Hardware on the order of **\$25–35k** for a single cell, dominated by the robot.

**Data + compute.** A BC baseline for one constrained task needs **~50–200 demonstrations** (~1–4 h capture at ~1 min/demo); diffusion wants more. Training a small BC/diffusion baseline is **a few GPU-hours to ~1 GPU-day** — cheap next to data collection, which is the real cost driver.

**Prototype → MVP timeline.** P0 (done): offline engine + sim replay. P1 (~1–2 wks): add depth/wrist-cam for object state. P2 (~1 wk): scale demos + LeRobot conversion. P3 (~1 wk): train + sim-eval a BC baseline. P4 (later): sim-to-real. **Tradeoff:** the RGB-only path is fast but caps at open-loop replay; depth + data is what unlocks a closed-loop policy. **Risks:** sim-to-real gap on a dexterous hand, object occlusion in egocentric views, and the data scale manipulation policies demand.

## 6. Limitations & next steps

Direct about what this does not do:

- **Egocentric depth is approximated** (no depth sensor): reach *direction* is right, absolute forward depth is a guess — still the biggest error source.
- **Left-hand coverage is low (26.5%)**, so bimanual moments still render one-handed.
- **Thumb-distance retarget is only reliable when the hand is fully in frame**; partial views fall back to curl.
- **No shared IMU ↔ video clock** → start offset is a best-effort assumption (~2–3 frames); IMU yaw is drift-leaked (no magnetometer).
- **Full-body pose is unrecoverable** from the forward cam (the founding observation), hence legs frozen + IMU-driven torso turn.
- **Inspire underactuation modeled as independent joints** (real hand ≈6 motors / 12 joints) — a deliberate simplification ( docs/inspire\_hand\_model.md ).
- **The real-time live web panel was designed but not built.** The concrete threaded architecture (native MuJoCo viewer + solver thread + uvicorn + MJPEG of the annotated video, frame-dropping to stay on the clock) is in docs/live\_architecture.md — verdict: feasible, biggest risk is Python GIL/CPU contention.

**Next steps:** add a depth / wrist-cam prior to fix reach depth; improve left-hand re-acquisition; detect a shared motion event to pin the IMU/video offset; collect demos and train the BC/diffusion baseline (§3); build the live panel per the architecture doc.

## 7. References

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- **MediaPipe Hands** — Zhang et al., 2020 (Google).
- **One Euro filter** — Casiez, Roussel, Vogel, CHI 2012, <https://gery.casiez.net/1euro/>
- **MuJoCo Menagerie / Unitree G1** — Google DeepMind, mujoco\_menagerie/unitree\_g1 .
- **Inspire hand description** — unitreerobotics/unitree\_ros , g1\_description/inspire\_hand (DFQ).
- **Damped-least-squares IK** — Buss & Kim 2005; Nakamura & Hanafusa 1986.
- **Imitation learning** — Diffusion Policy (Chi et al., 2023); ACT/ALOHA (Zhao et al., 2023); OpenVLA (Kim et al., 2024) /  $\pi$ -0 (Physical Intelligence, 2024); LeRobot (Hugging Face) dataset format; AnyTeleop dexterous retargeting (Qin et al., 2023).
- **Egocentric full-body limitation** — e.g. Tome et al., "xR-EgoPose", ICCV 2019.
- Project docs: docs/diagnostics.md , docs/inspire\_hand\_model.md , docs/live\_architecture.md .